Unit 2 - Serverless Architecture

What is Serverless?

Serverless computing is a cloud execution model where the cloud provider dynamically manages the allocation and provisioning of servers. Despite the name, servers still exist - developers just do not manage them. You only pay for actual execution time, not for idle server time.

AWS Lambda

AWS Lambda is the core of serverless on AWS. It runs code in response to events and automatically manages compute resources. Lambda supports Python, Node.js, Java, Go, and more. Functions are triggered by events from S3, DynamoDB, API Gateway, Cognito, and many other services.

Lambda Key Concepts

Function

The code you run on Lambda. Each function has a handler, runtime, memory, and timeout setting.

Event

The trigger that invokes the Lambda function. Can be HTTP request, file upload, database change, schedule, etc.

Execution Role

IAM role that grants Lambda permissions to access other AWS services like DynamoDB and S3.

Cold Start

Initial latency when a Lambda function is invoked for the first time or after being idle. Can be mitigated with Provisioned Concurrency.

Serverless Architecture Pattern

- Client sends request to CloudFront (CDN)
- CloudFront serves static files from S3
- Dynamic requests go to API Gateway
- API Gateway triggers Lambda functions
- Lambda reads/writes to DynamoDB
- Lambda can call external services like Cognito
- Response flows back through API Gateway to client

Benefits of Serverless

- No server management required
- Automatic scaling from zero to millions of requests
- Pay only for what you use (per millisecond billing)
- Built-in high availability across multiple AZs
- Faster time to market for new features
- Reduced operational overhead

AWS DynamoDB

DynamoDB is a fully managed NoSQL database service. It provides fast and predictable performance with seamless scalability. It stores data as items (rows) in tables with a primary key. DynamoDB supports single-digit millisecond response times and can handle millions of requests per second.